Type MS · Draw or label a human cell as seen under the microscope

Answer 1 of 6 · Question 2 part a · 4 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

2 (a)		• irregular shaped cell; • nucleus labelled; • cytoplasm labelled; • cell/plasma membrane labelled;		4
(b) (i)		• similar cells/cells with similar characteristics/shape; • grouped/work together; • to carry out a particular function;		3
	(ii)	squamous epithelium;		1
	(iii)	any two from • nervous; • ciliated epithelium • blood; • bone; • (smooth/cardiac/skeletal) muscle; • connective; • skin	reject 'a' bone' reject 'a' muscle'	2

Total 10 marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 1 of 6 · Question 2 part a · 4 marks

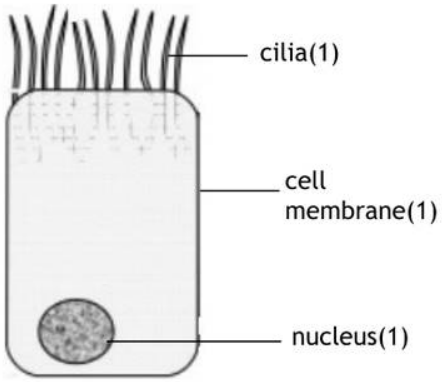
2023 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 2 of 6 · Question 2 part a.i · 3 marks

2020 NOV · Paper 01 (regional R) · find part a.i in this mark scheme extract

2 (a) (i)			3
(ii)	holds the genetic code cilia site where most chemical reactions occur cell membrane controls entry and exit of makes proteins nucleus moves mucus through the trachea	Reject more than one line from each structure	3
(b) (i)	Tar (1)		1
(ii)	• (cilia)burned/destroyed/reduced in number(1) • paralysed/cannot beat to and fro/move mucus(1)		2

Total for Question 2 = 9 Marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 3 of 6 · Question 8 part a.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

8 (a) (i)	chromosomes (1)			1
(ii)	 (1)  (1)			2
(b) (i)	complementary bases pair up 3 DNA double helix unwinds 1 strands separate 2 two DNA strands form 4			1
(c) (i)	thymine/T (1)	R thiamine/thyamine		1
(ii)	guanine/G/cytosine/C (1)			1
(c)	any four from • allows formation of gametes (1) • haploid (1) • allows variation to occur (1) • which allows species to evolve (1) • allows diploid number to be maintained (1) • at fertilisation (1)			4

Total 10 marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 3 of 6 · Question 8 part a.i · 1 mark

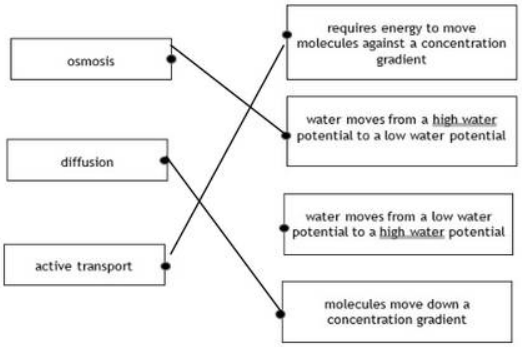
2020 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Draw or label a human cell as seen under the microscope

Answer 4 of 6 · Question 1 part a.i · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Question number				
1	(a) (i)	X = cell membrane; Y = nucleus; Z = cytoplasm;	Allow nuclear membrane	3
	(ii)	mitochondria/ribosomes/endoplasmic reticulum		1
	(iii)	• similar cells work together to form tissues; • (different) tissues work together to form organs;		2
	(b)		Reject more than one line from each process	3
	(c) (i)	A = squamous (epithelium) B = ciliated (epithelium)		2
	(ii)	60 x 40; 2400 (µm) (÷1000); 2.4 (mm);	Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015	3

Total 14 marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 4 of 6 · Question 1 part a.i · 3 marks

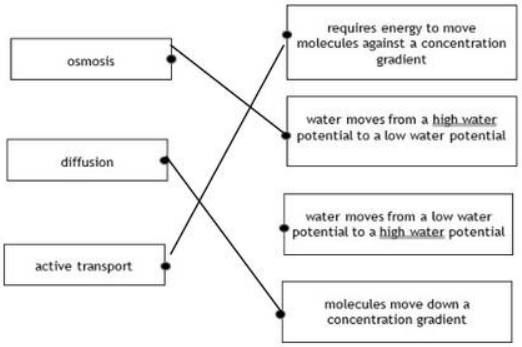
2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 5 of 6 · Question 1 part a.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
1	(a) (i)	X = cell membrane; Y = nucleus; Z = cytoplasm;	Allow nuclear membrane	3
	(ii)	mitochondria/ribosomes/endoplasmic reticulum		1
	(iii)	• similar cells work together to form tissues; • (different) tissues work together to form organs;		2
	(b)		Reject more than one line from each process	3
	(c) (i)	A = squamous (epithelium) B = ciliated (epithelium)		2
	(ii)	60 x 40; 2400 (µm) (÷1000); 2.4 (mm);	Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015	3

Total 14 marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 5 of 6 · Question 1 part a.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw or label a human cell as seen under the microscope

Answer 6 of 6 · Question 6 part a · 4 marks

2023 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

6 (a)	• general shape with nucleus at centre; • nucleus labelled; • cytoplasm labelled; • cell membrane labelled;		4
(b)	• scrape/swab cells from (inside of) cheek • smear (thinly) over slide;/place sample on slide; • add stain/cover with coverslip;		3
(c)	• cheek cell shape irregular, trachea cell oblong; • cilia present in trachea, absent in cheek cell; • position of nucleus different;		3

Total 10 marks

Question number	Answer	Notes	Marks